



Developing a satellite-based combined land degradation index: a case study in Tana-Beles watershed, Upper Blue Nile, Ethiopia

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Nov 11, 2022

Addis Ababa, Ethiopia



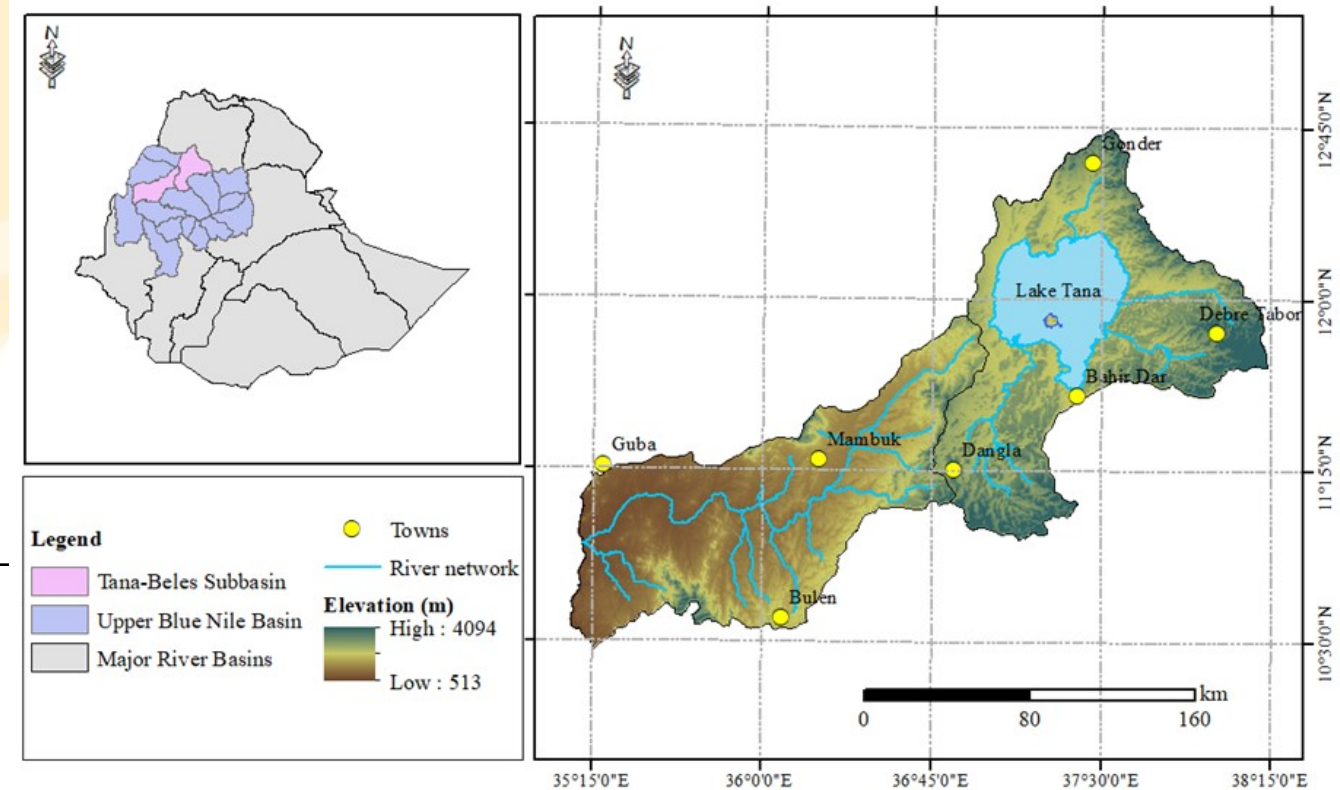
Developing a satellite-based combined land degradation index: a case study in Tana-Beles watershed, Upper Blue Nile, Ethiopia

Introduction

- Land degradation (LD) is a negative trend in land conditions;
- LD is worsening in many regions: developing countries are highly affected
- Early identification of land degradation is particularly important for countries like Ethiopia, where semi-arid and arid climate zones occupy 60% (Asfaw et al., 2018);
- The study focuses on developing a comprehensive combined land degradation index in the Tana-Beles watershed using remotely sensed data (Landsat image);
- The study also evaluates the environmental change associated with land degradation over the study area;

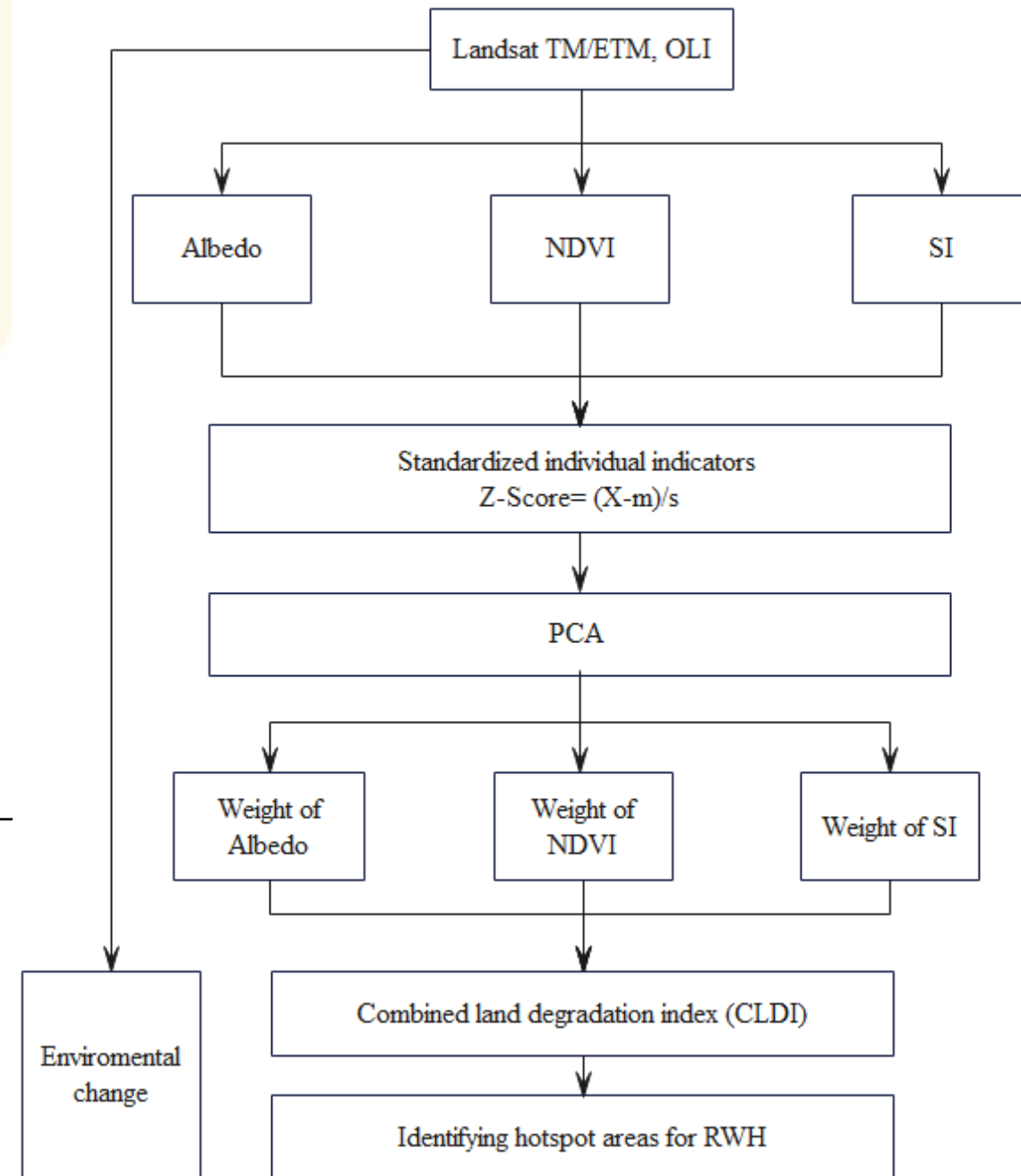
Study area

- Sub watersheds of Upper Blue Nile Basin;
- Geographically, the Tana-Beles watershed is located at 10°22'N, 35°03'E and 12°50'N, 38°21'E
- Lake Tana and Beles watersheds: connected through a tunnel for inter-basin water transfer since May 2010.
- Area 29,260 km²
- Considered as one of Ethiopia's agricultural growth corridors



Methods

- Combines multiple LD indicators by applying Principal Component Analysis (PCA)
- Vegetation condition (normalized difference vegetation index - NDVI)
- Soil health: soil salinity (salinity index-SI)
- Soil exposure (soil albedo);
- Landsat 5 and 8 were used for the period 1990 to 2000 (the 1990s) and 2010 to 2020 (2010s),
- For the 1990s, there were 1,255 Landsat images, and for the 2010s, there were 1,284 images to estimate (Albedo, NDVI, and SI)
- Finally, identifies hotspot areas for RWH structure implementation.
- Google Earth Engine was used for data processing



Methods...

$$\text{Albedo (TM)} = 0.356\rho_{\text{Blue}} + 0.130\rho_{\text{Red}} + 0.373\rho_{\text{NIR}} + 0.085\rho_{\text{SWIR}} + 0.072\rho_{\text{MIR}} - 0.0018$$

Albedo [0.1 to 0.8, forest and fresh snow] (Liang, 2001; Naegeli et al., 2017)

$$\text{NDVI} = \frac{\text{NIR} - \text{Red}}{\text{NIR} + \text{Red}}$$

NDVI [-1 to 1, > 0.6 dense forests, < 0 waterbodies]

$$\text{SI} = \sqrt{\text{Blue} * \text{Red}} \quad (\text{Ennaji et al., 2018})$$

SI [0 to 1, < 0.4 considered as none saline soil]

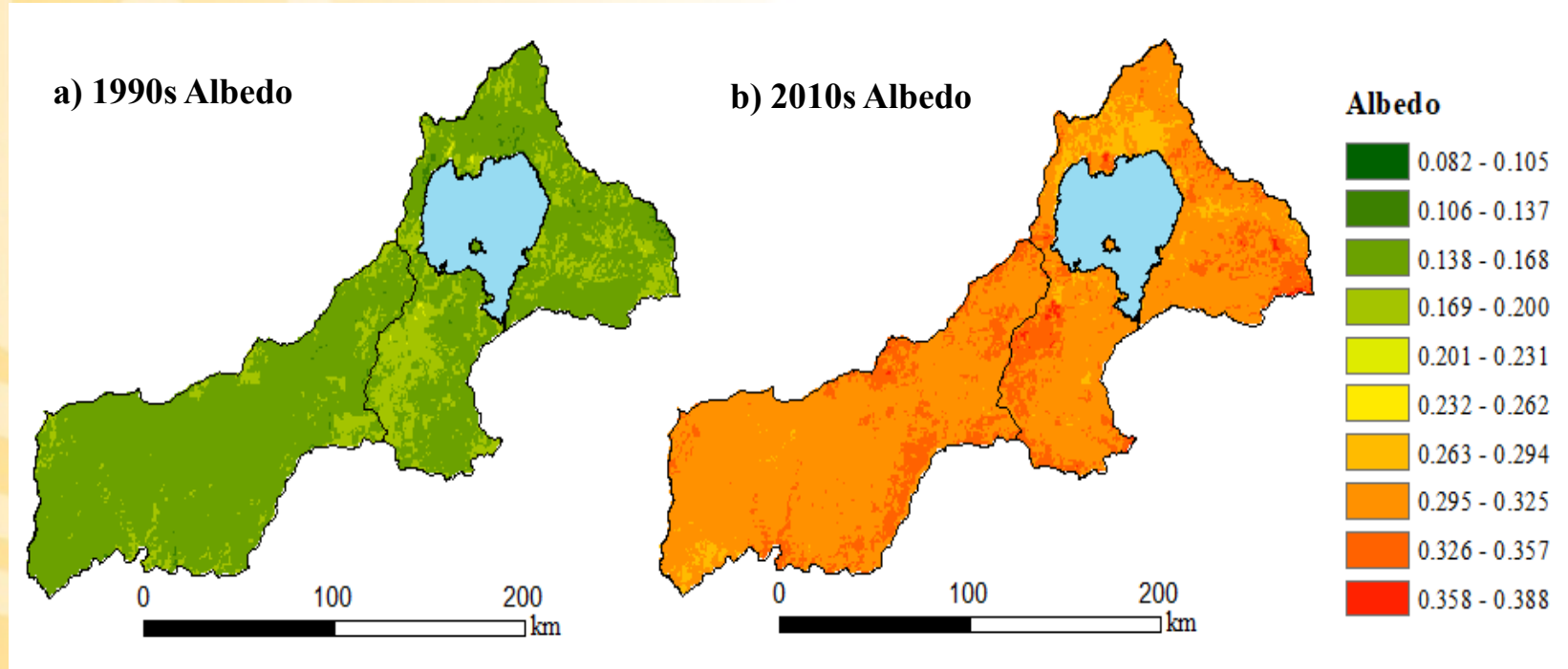
Combined Land Degradation Index (CLDI)

$$\text{CLDI} = w_{\text{albedo}} * \text{Albedo} + w_{\text{NDVI}} * \text{NDVI} + w_{\text{SI}} * \text{SI}$$

Environmental change: land use classification and water quality

Results

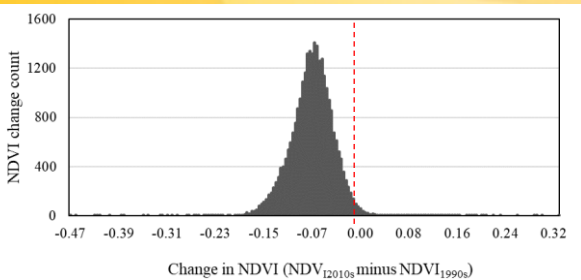
Albedo



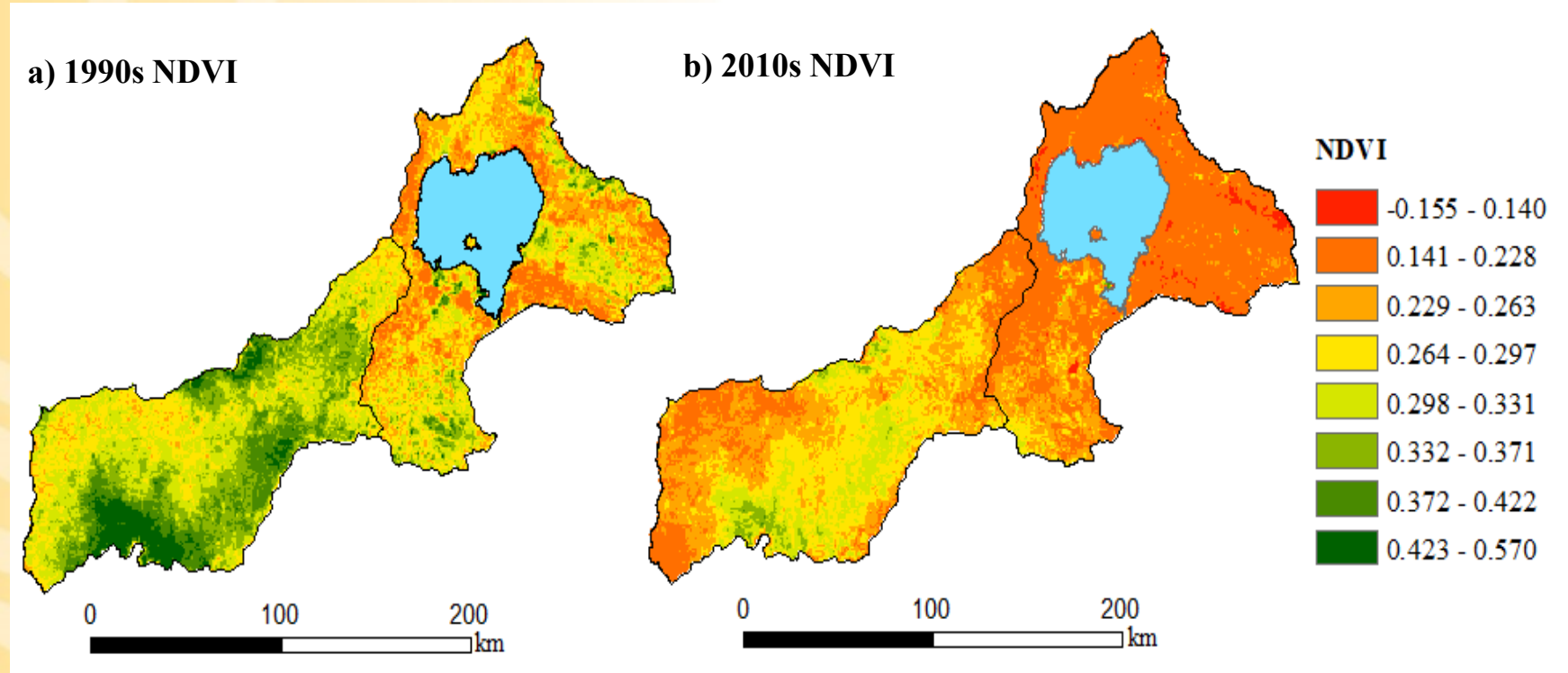
- Soil Albedo increased by 94% in the 2010s compared to the 1990s;
- Increased albedo will have a warming effect and also affects evaporation since it partitions sensible and latent heat flux.

Results...

NDVI



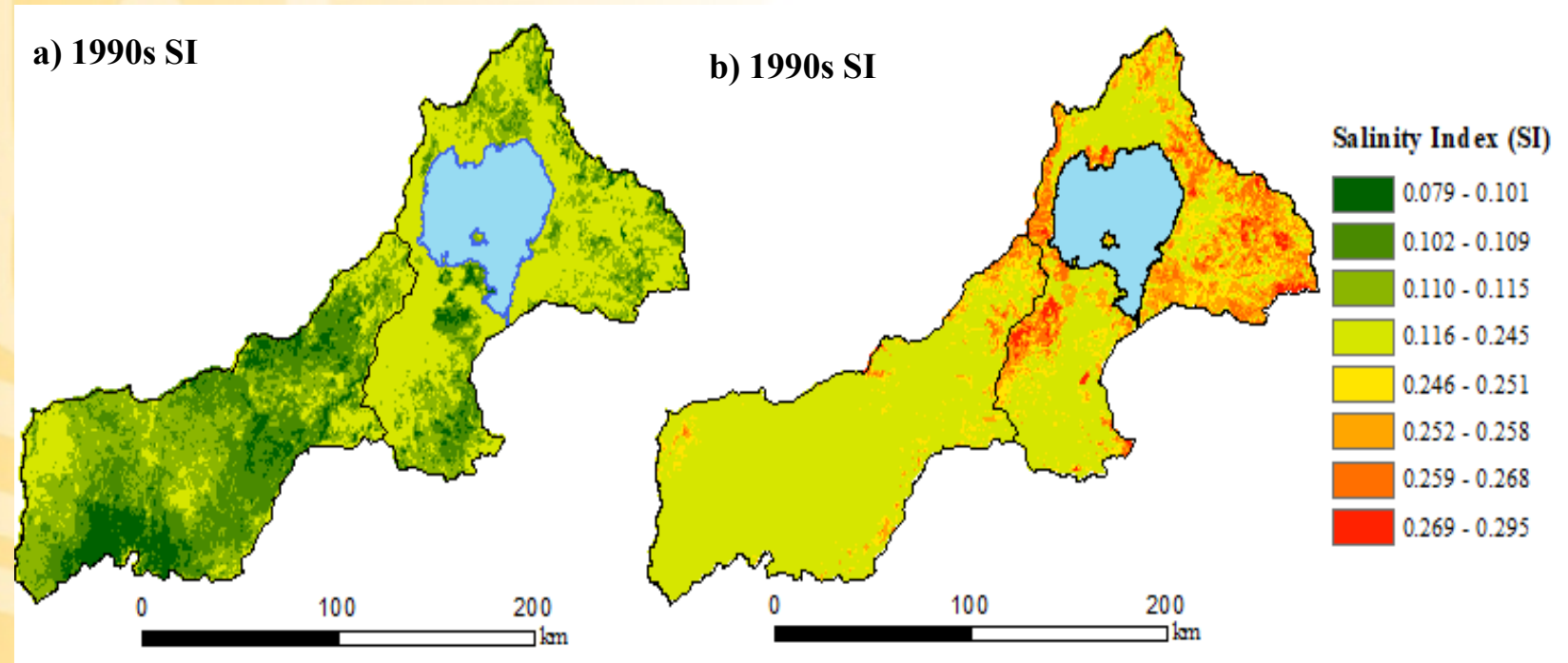
Histogram of the change in NDVI



- NDVI 1990s ~ 0.305; NDVI 2010s ~0.232
- NDVI decreased by 23% in the 2010s compared to the 1990s
- Almost in 98.5% of the area, vegetation conditions deteriorated;

Results...

Salinity (SI)



- SI 1990s ~ 0.113 and 2010s ~ 0.205
- Soil salinity index increased by 82% in the 2010s compared to the 1990s;
- However, the soil salinity in 2010s is considered as non-saline since the salinity index is less than 0.4 or 2Ece;

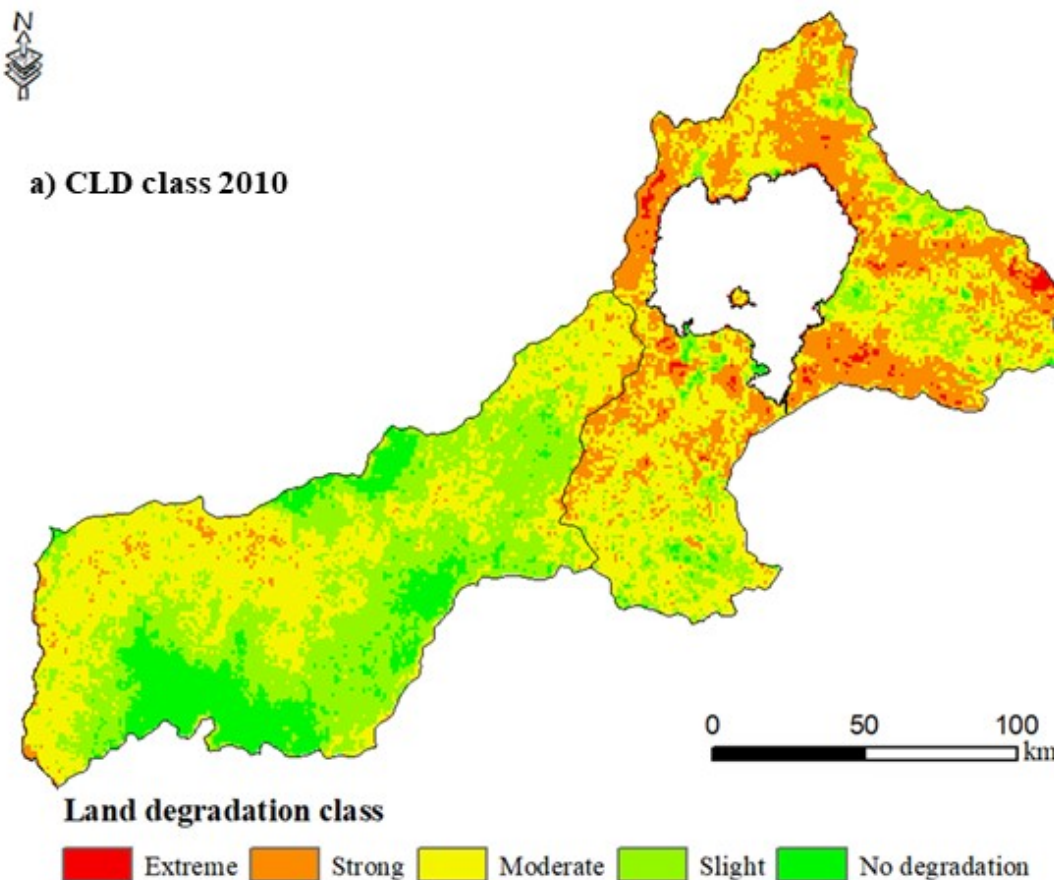
Results...

CLDI

Land degradation class (2010s)	Fraction of area (%)
Extremely degraded	2
Strongly degraded	18
Moderately degraded	43
Slightly degraded	27
Not degraded	10



a) CLD class 2010



- PCA was applied to reduce the dimensionality to develop CLDI
- Weights of individual indexes were determined based on the variance of the data.

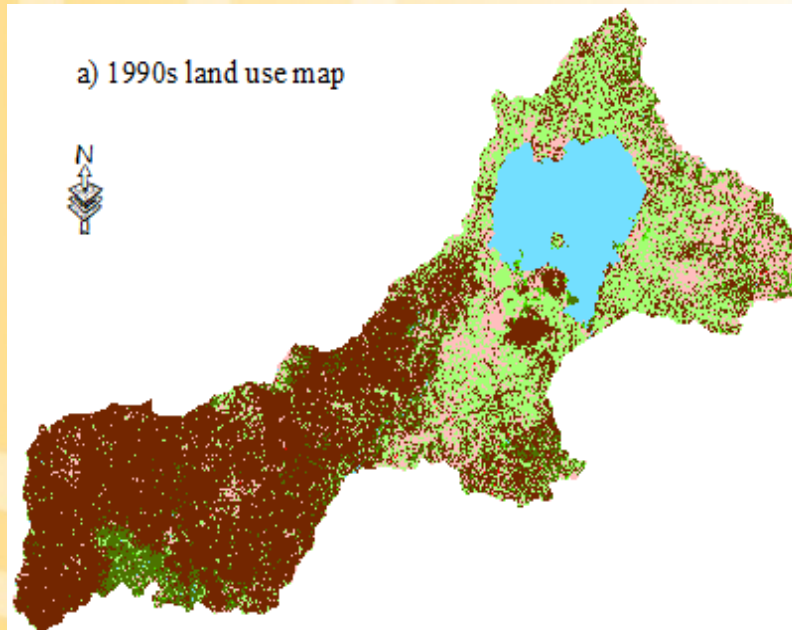
$$CLDI = 0.085 * lbedo + 0.91 * NDVI + 0.005 * SI$$

Results...

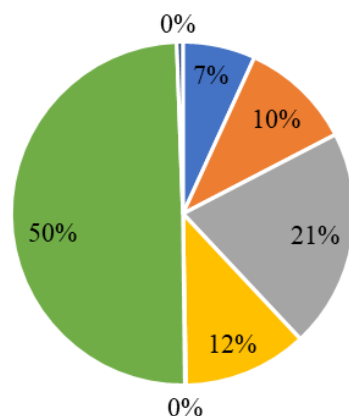
Environmental change

- Supervised landuse classification

Landuse	Acc. Asses.
1990s	95%
2010s	96%

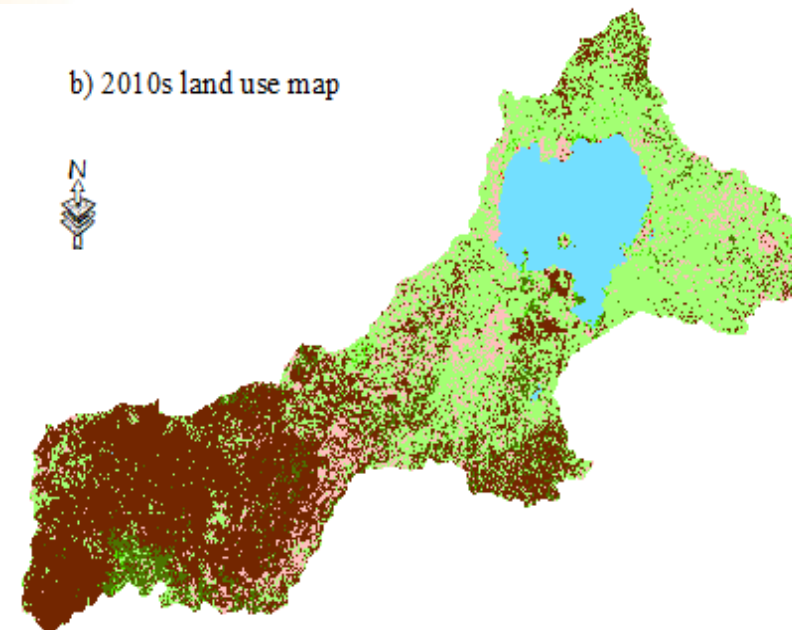


0 80 160 km

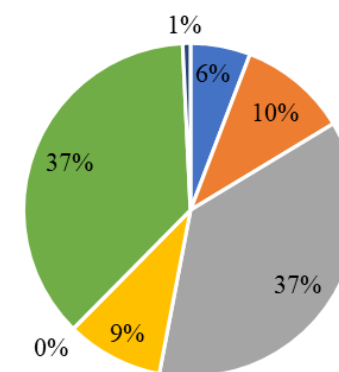


Landuse 1990s

Forest Water body Cropland
Bareland Settlement Bushland
Wetland



0 80 160 km



Landuse 2010s

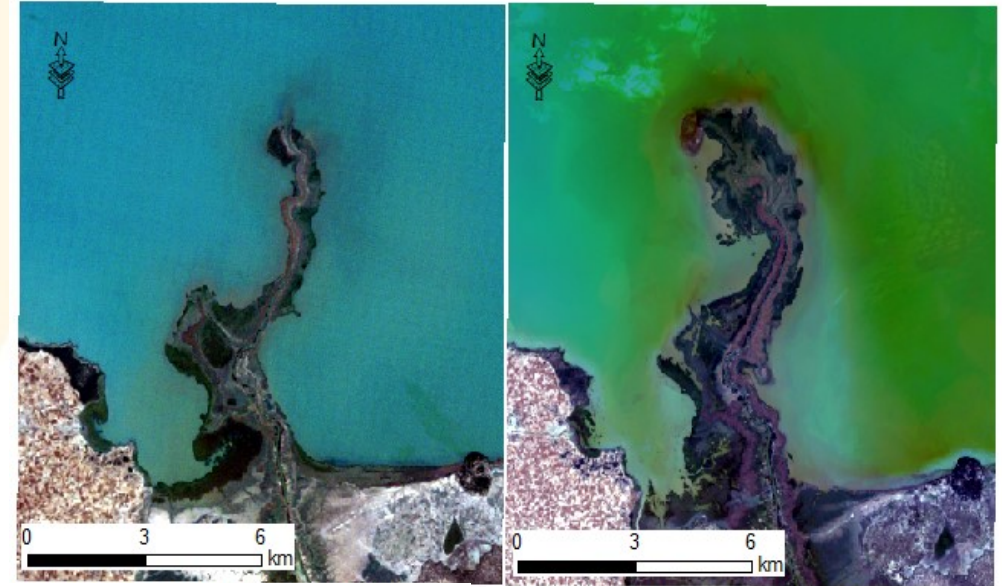
Forest Water body Cropland
Settlement Bushland Wetland
Bareland

Landuse

Bareland
Bushland
Cropland
Forest
Settlement
Water body
Wetland

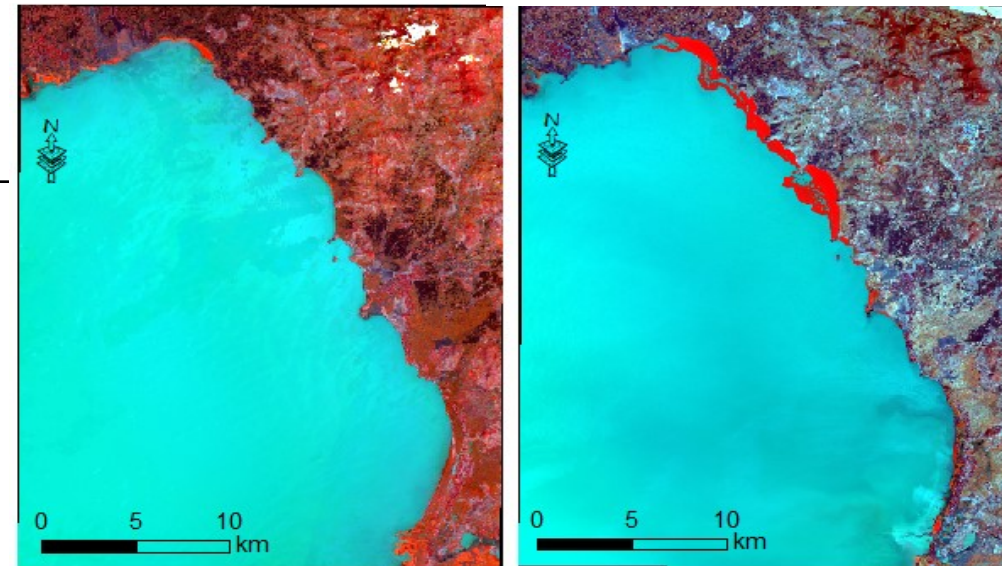
Environmental change

- Soil erosion in the head watershed: Gilgel Abay River mouth delta expanded considerably due to sediment deposition between the 1990s and 2010s. River mouth delta increased by 49% (i.e., from 14.6 km² to 21.8 km²).
- The sediment coming with the surface runoff bringing along nutrients has caused eutrophication of the lake, making the lake a fertile ground for water hyacinth;
- The water hyacinth area increased by 700% in 2020 compared to 2016 (i.e., from 2.5 km² ha to 20 km²).



a) 1990s river mouth delta

b) 2010s River mouth delta

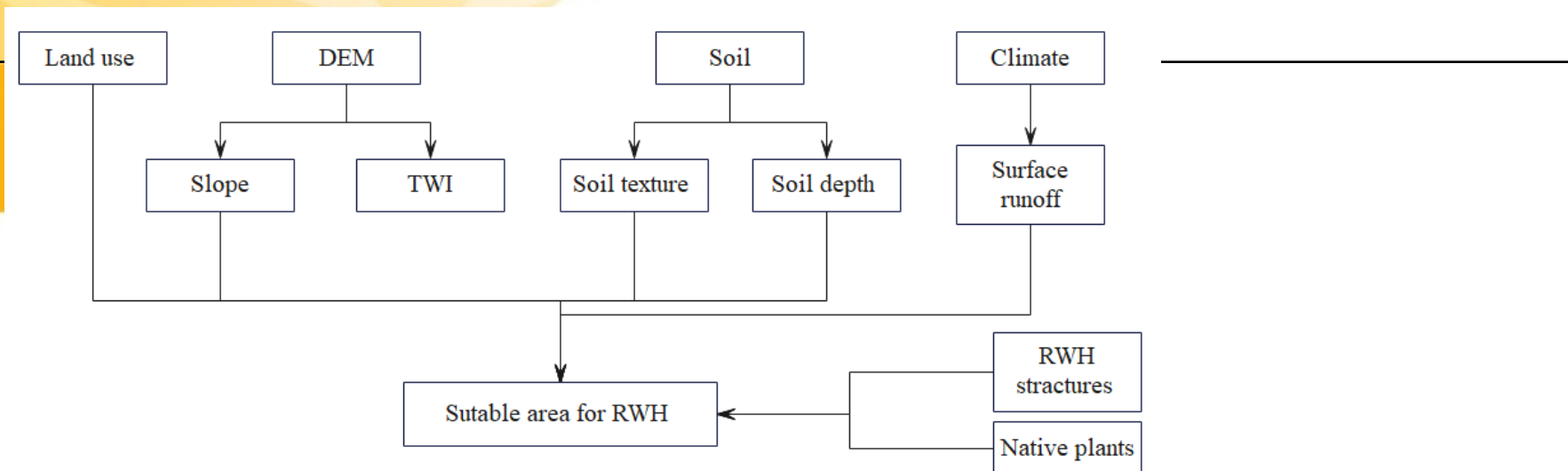


a) 2016 Water hyacinth

b) 2020 Water hyacinth

Linkage with RWH

- The study identifies the spatial location of degraded areas across the Tana-Beles Basin;
- LD assessment can be used to prioritize interventions to reduce land degradation;
- Contraction of RWH (BMPs) combined with plantation of native species can potentially restore the LD in Tana-Beles;
- A GIS-based Multi-Criteria Evaluation (MCE) technique can be applied to identify a potential area for the native plants (shrubs) and strictures;



Conclusions

- The study combined multiple land degradation indicators gauging vegetation condition, soil health, and soil exposure;
 - Land degradation is the major issue in the Tana-Beles Basin;
 - ~63% of Tana-Beles is estimated to be moderately to extremely degraded;
-
- Huge environmental change due to land use change and soil erosion.
 - To reverse LD in the basin, putting in place RWH structures and planting native plants will provide an opportunity to stabilize the landscape and build resilience;

Thank you

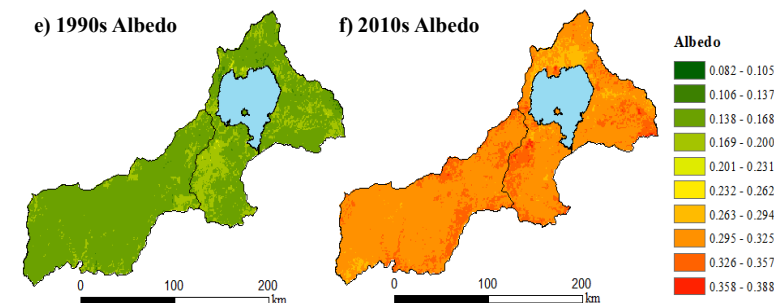
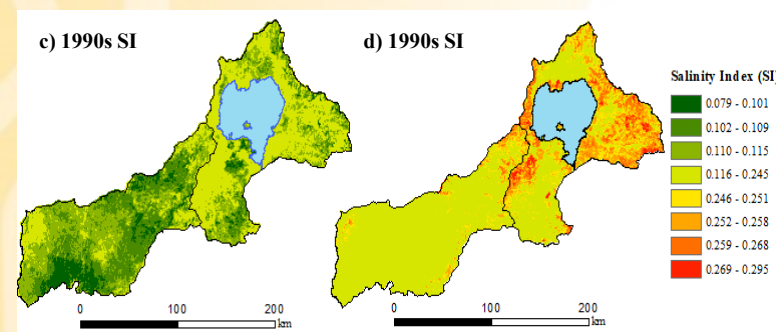
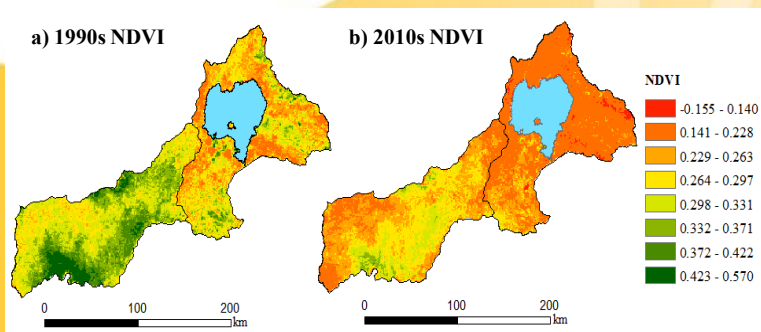
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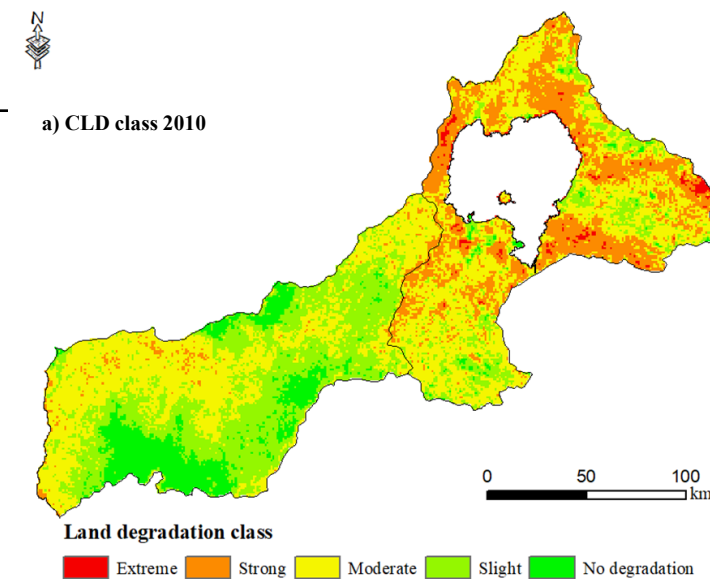
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